thepulse-markets

A Common Lisp Client for the Pulse Verity Index

https://github.com/cgore/cl-thepulse-markets/

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Introduction

`thepulse-markets` is a Common Lisp client for the Pulse Verity Index, the read API documented at `https://thepulse.markets/developers`. It is not an official Pulse product. The calls go to

`https://mcp.thepulse.markets/api/index/v1/`

Pulse publishes a USD index for crypto tickers. A price response carries an ECDSA P-256 signature over four fields: the symbol, the price exactly as text, the timestamp, and the grade. The library checks that signature locally against the public key ring. It does not ask Pulse whether a stored print is still good.

The thin layer lives in `thepulse-markets/rest-api`: bearer auth, 429 retries that stop for a monthly cap, JSON via Yason, and one function per documented REST read. The `thepulse-markets` package re-exports that layer and adds the helpers you actually want: `price`, `verify-print`, `ohlc-history` as a list of `ohlc-bar` objects, and `ohlc-bars-for-ingest` for candlesticks [1].

The shape follows `cl-coingecko` [2] and `yahoo-finance` [3]. Specs are `sigma/behave` forms and run when the system loads.

The WebSocket stream is not wrapped. The developer console (keys, billing, webhooks) is a browser session on `thepulse.markets`, not this API key, and is not wrapped either.

Chapter 1

The REST Client

The HTTP layer. It sends `Authorization: Bearer` only on keyed paths, refuses redirects, retries HTTP 429 `RATE_LIMITED`, and does not retry `MONTHLY_LIMIT`. JSON arrays become vectors and booleans become symbols. A body with `success: false` is an `api-error` even when the status was 200.

1.1 Variables

1.1.1 The `*api*` Variable

`https://mcp.thepulse.markets`. Paths hang off `/api/index/v1/`.

1.1.2 The `*api-key*` Variable

The developer key. On load, the gitignored `local-configuration.lisp` file is loaded when it exists and may rebind this. Otherwise the library reads `PULSE_API_KEY`, then `THEPULSE_MARKETS_API_KEY`. The file wins over the environment. `use-key` and `use-keyless` set it in the image. Placeholders such as `pidx_your_key_here` are treated as no key.

1.1.3 The `*api-cache-timeout-seconds*` Variable

How long `http-get-json-cached` keeps a response. Default 60. Live price calls default to `:cache nil`. The cache is partitioned by a short hash of the key, not by the key itself.

1.1.4 The `*pubkey-cache-timeout-seconds*` Variable

How long `get-pubkey` keeps the verification ring. Default 300.

1.1.5 The `*max-retries*` Variable

Attempts for a rate-limited response. Default 3. Attempt n waits n times `*retry-wait-seconds*` (default 2), or `Retry-After` when that header is between 1 and 30 seconds.

1.2 Conditions

All of these are subclasses of `api-error`. Readers are `api-error-status`, `api-error-url`, `api-error-code`, `api-error-message`, and `api-error-body`.

`rate-limited`	HTTP 429, wait and retry
`monthly-limit`	HTTP 429, do not retry
`unauthorized`	HTTP 401
`plan-restricted`	HTTP 403
`unsupported-symbol`	the ticker is not served
`service-unavailable`	HTTP 503
`needs-key`	a keyed call with no usable key

An unsupported symbol on `/price` signals. The same symbol inside `get-batch` is one row with `success` false. It is not a price of zero.

1.3 Symbols and time

1.3.1 The `canonicalize-symbol` Function

Uppercase a ticker. `:btc` and `"btc"` are both `"BTC"`.

1.3.2 The `canonicalize-symbols` Function

A symbol, a comma string, or a list. Duplicates drop, order stays.

1.3.3 The `canonicalize-interval` Function

`1m`, `5m`, `1h`, `1d`. Short names `m`, `h`, and `d` map onto those. The Lisp reader upcases, so `:m` and `:M` are the same symbol and mean a minute.

1.3.4 The `canonicalize-time` Function

An ISO-8601 string passes through. An integer above 10^{11} is UNIX milliseconds, above $3 \cdot 10^9$ is a universal time, and anything smaller is UNIX seconds. The result is `YYYY-MM-DDTHH:MM:SSZ`.

1.3.5 The `parse-iso-8601` Function

The inverse, to universal time. Fractional seconds are truncated. `Z` and numeric offsets are accepted.

1.4 Reads

Each `get-*` function is the REST path of the same name. `get-price` is `/price` and always requires a key. `get-sample` is the keyless BTC, ETH, and SOL print. `get-batch` accepts up to 100 symbols. `get-print` is the signed print nearest a time, plus `deltaMs`. `get-history` takes `:dataset verified` or `reconstructed`, an interval, and an optional page `:cursor`. `get-research-compare` and `get-source-health` are the plangated research reads. `get-verity-price`, `get-verity-batch`, and `get-verity-catalog` are the unsigned breadth surface. Public, and not sent the key: `get-pubkey`, `get-symbols`, `get-health`, `get-status`, `get-verity-status`, and `get-verity-venues`.

Results are Yason hash tables. Pass `:cache nil` to skip the cache. `get-pubkey` uses its own five-minute cache.

Chapter 2

Using the Library

The `thepulse-markets` package is what application code uses. It re-exports the REST client and adds prices, local signature checks, and OHLC history.

2.1 Prices

2.1.1 The `price` Function

Current index price. The primary value is a number. The second value is the signed print.

With a usable key this is `get-price`. Without one, `:btc`, `:eth`, and `:sol` come from the sample and anything else signals `needs-key`. `btc`, `eth`, and `sol` are those three calls.

`:cache` defaults to nil. A 60-second cache would hand back a stale signed print.

2.1.2 The `prices` Function

One plist per symbol: `:symbol`, `:price`, `:print`, `:code`, `:message`. One `/batch` call when a key is set. An unknown ticker is a row, not a failed batch.

2.1.3 The `settlement-print` Function

The signed print nearest a time. Values are the price, the print, and `deltaMs`. `deltaMs` is the distance from the time you asked for. It is not signed. The print is the nearest recorded sample, not a price synthesized at that instant.

2.1.4 The `verity-price` Function

The unsigned breadth read. `verify-print` does not apply.

2.2 Verification

The v1 canonical string is five lines:

```
pulse-index-v1
symbol
price text
timestamp
grade
```

The price text is the `priceText` field when Pulse sent one. Reprinting the float is a fallback and can fail a check that the original text would pass. If `price` and `priceText` are different numbers, `verify-print` returns nil with reason `:price-text-mismatch` and does not treat the print as merely badly signed.

The signature is ECDSA P-256 with SHA-256, IEEE-P1363 `r||s`, standard base64. The public key is the uncompressed P-256 point from the published JWK, or the tail of the PEM. The ring is at most 16 keys. A `kid` selects that key. A print with no `kid` is tried against the ring.

2.2.1 The `verify-print` Function

Primary value is true or false. The second value is a plist: `:valid`, `:v1`, `:v2`, `:kid`, `:checked`, `:reason`.

`:version :v1` is the default. `:v2` checks the canonical string Pulse stored under `v2`, which includes tier, confidence, cadence, and the interval. Do not rebuild that string. `:both` requires v1, and requires v2 when the block is present. A print that has only v1 still passes `:both`.

With no `:key-ring` argument the ring is fetched, and fetched once more if the signature fails. A ring you pass is not refreshed. True does not authenticate fields outside the canonical string you checked.

2.3 History

2.3.1 The `ohlc-history` Function

A list of `ohlc-bar` and a meta plist. The default window is the last 24 hours. `get-history` with no bounds walks from the start of the archive; this function does not. The default interval is one minute.

Verified history rebuilds bars from the signed observations, so each interval in the window is one bar even when the read spans pages. The reconstructed dataset has candles only. Fragments that share a timestamp are merged: open from the first, close from the last, high the max, low the min. Reconstructed history is not a contemporaneous signature. The meta plist says which, in `:contemporaneously-signed`.

`:limit` requests one page. `:follow t` walks cursors, stopping after `*history-max-pages*` (20). `:truncated` is true when the walk stopped early.

Volume and adjusted close are nil. Pulse does not report trade volume.

2.3.2 The `price-history` Function

`(universal-time price)` pairs and the same meta plist. Verified history returns every observation. Reconstructed history returns the candle close. Universal time drops the fractional second.

2.3.3 The `ohlc-bars-for-ingest` Function

The tuple list `candlesticks:ingest-candlesticks` accepts: `(time open high low close volume adjusted-close)`. This library does not depend on candlesticks. The quote currency in the meta plist is USD.

2.3.4 The `ohlc-bar` Type

Slots `time`, `open`, `high`, `low`, `close`, `volume`, and `adjusted-close`, with accessors `ohlc-bar-time` and so on. `time` is a universal time.

Bibliography

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[2] Gore, C. M. cl-coingecko. https://github.com/cgore/cl-coingecko, 2026.

[3] Gore, C. M. yahoo-finance. https://github.com/cgore/yahoo-finance, 2026.